Moeiz Shahzad

Senior AI Engineer

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SUMMARY

AI Engineer with 4+ years building production AI and backend systems in Python. Designs multi-agent automation that plans, builds, tests and self-repairs workflows from plain-language requests, connected to real business tools through MCP and running on durable Temporal execution. Combines distributed backend architecture (FastAPI, Django, Temporal) with applied ML, computer vision and satellite remote sensing, and has taken AI products from research to enterprise production. Has led a team of 10+ engineers and mentored nearly 20.

TECHNICAL SKILLS

AI & LLM: AI agents, agentic workflows and automation, multi-agent systems, LLMs, generative AI, NLP, tool/function calling, structured outputs, prompt and context engineering, MCP, AutoGen, LangChain/LangGraph, LlamaIndex, LiteLLM, OpenAI, Claude, Gemini, LLaMA

RAG & Retrieval: RAG, agentic RAG, GraphRAG, hybrid search (dense + BM25), reranking, chunking strategies, text-to-SQL, multimodal RAG, OCR and document AI, embeddings, Pinecone, Chroma

LLM Engineering & Evaluation: fine-tuning (SFT, LoRA), quantization, guardrails (prompt-injection protection), evaluation (intent-based, LLM and SLM judges), agent performance tracking and observability, semantic caching, Redis caching, model routing and fallbacks, streaming, latency and cost optimization

Backend & Distributed Systems: Python, FastAPI, Django, Django REST Framework, Blacksheep, Node.js, Express.js, Temporal, Apache Kafka, Celery, AsyncIO, microservices, distributed systems, REST APIs, system architecture

Data & Cloud: PostgreSQL, MongoDB, Redis, ClickHouse, MySQL, MariaDB, Docker, Kubernetes, AWS, Google Cloud, Linux (Ubuntu), GitHub Actions, CI/CD

ML & Computer Vision: Machine learning, deep learning, XGBoost, Random Forest, Scikit-Learn, TensorFlow, CNNs, YOLO, OpenCV, NumPy, Pandas, digital image processing, computer vision, time-series and regression analysis

Geospatial and Integrations: Google Earth Engine, Sentinel Hub, remote sensing, satellite imagery, Leaflet, Mapbox, smart agriculture. **Integrations:** Stripe, Healthie EHR, Stedi, Frappe/ERPNext, ERP modules, browser extensions

Languages & Practices: Python, TypeScript, JavaScript, SQL, React.js, Next.js, React Native, System Design, Agile/Scrum, Jira, Claude Code, Cursor AI

EXPERIENCE

Senior AI Engineer · Devsinc — Aug 2025 – Present

- Designed and built a **multi-agent automation system** (orchestrator, tool-discovery, connection and code-synthesis agents) that turns plain-language requests into tested, working automations, an AI-native alternative to Zapier and n8n.
- Built a test, judge and repair loop that checks each generated workflow against the user's intent, fixes logic errors automatically and escalates only credential or platform issues to a human. A correct workflow is ready in **2 to 3 iterations** (4 to 5 for complex 40 to 50 node workflows, tested in batches).
- Integrated **MCP servers** and REST tools with dynamic tool discovery, injecting credentials at runtime so generated code running in isolated sandboxes never sees them.
- Added guardrails across the builder's collaborating agents with Markdown and Jinja templates that clearly separate system context from user input, so user text cannot rewrite an agent's role or push it off task.
- Built the durable execution layer on **Temporal** (replayable history, automatic retries, human-approval waits) so long-running agent workflows survive crashes and resume exactly where they stopped, serving about 1,100 users and **400,000+ workflow runs** (builder runs plus the workflows it generates).
- Optimized the prompt-to-workflow pipeline for dynamic task referencing and context integrity, cutting LLM operating costs by about **35%** in customer-facing production.
- Gave each agent role its own LLM provider with a same-family backup for consistent failover, tracked per-agent performance, call counts and context-window overflows on dev and stage, and added semantic caching that starts the builder from a workflow matching about 70% of a new request.

Software Engineer · Gicoh — Feb 2024 – Sep 2025

- Delivered enterprise backend systems end to end on **Frappe/ERPNext**, building and maintaining HR, Manufacturing, Accounting, CRM, Inventory, Procurement and Supply Chain modules from requirements through production.
- Optimized MySQL/MariaDB schemas and indexing for high-volume transactional workloads, with secure, high-concurrency data handling.

Backend Engineer · Nexlify Solutions — Jan 2023 – Aug 2025

- Led **CropSight** from a university research project to a production precision-agriculture platform serving sugar and wheat mills, serving **130+ clients**, 7 to 8 mills and 1,200+ users, and managing a cross-functional team of **11** from requirements to delivery.
- Owned infrastructure and DevOps end to end (physical servers, Nginx, SSL, automated deployments), sustaining **99.9% availability**.

Freelance Software & ML Engineer · Remote clients — Feb 2022 – Sep 2023

- Delivered Django/Flask backends and React/MERN applications for clients in inventory, HR and finance, plus ML models and data pipelines in Python and Scikit-Learn.

PROJECTS

VortexAI · Autonomous Workflow & Agent Builder Engine — Aug 2025 – Present

- Built a dual-evaluator simulation engine: workflows run in isolated test patches, where small language models check syntax and evaluator agents score completion against the user's intent.
- Unified LLM calls across OpenAI, Claude, Gemini and local models with **LiteLLM**, and solved AutoGen context-window overflow with custom state compression and windowing.
- Architected a native code-generation agent that writes multi-step workflows and standalone agents, enforcing platform design language, error handling and credential-management standards.

CropSight · Satellite Precision-Agriculture Platform — 2023 – 2025

- Built a farm-management platform that turns satellite imagery into crop-health monitoring, yield forecasting and fertilizer and irrigation optimization across **70,000+ hectares** at provincial scale.
- Engineered an automated remote-sensing pipeline on Google Earth Engine and Sentinel Hub: NDVI/EVI health indices, time-series analysis, field-level growth tracking and weather-soil data fusion.
- Trained an LSTM yield model on Sentinel-2, MODIS and Landsat time series (**RMSE 1.4 t/ha, R^2 0.93**) and deployed Scikit-Learn and TensorFlow models for sugarcane and wheat classification.
- Shipped the full product: Django and FastAPI backend, Next.js and React web dashboards and a React Native mobile app covering field management, task scheduling, job tracking and compliance.
- Cut repeat satellite lookups with Redis caching: authentication and live field-health data are cached with a 3-day TTL matched to the imagery refresh cycle, so views inside that window are cache hits instead of Sentinel Hub calls.

ClinicalAI EHR · Node.js, PostgreSQL, GraphQL, OpenAI — Aug – Nov 2025

- Built a real-time doctor portal with WebSocket audio streaming, **OpenAI Whisper** transcription and LLM pipelines that generate clinical summaries and prescriptions written straight back to the EHR, with about 1,000 concurrent live patient sessions.
- Integrated Healthie EHR over GraphQL as the source of truth, syncing up to **10,000 appointments**, patient records and practice structures into PostgreSQL.
- Delivered a patient portal and billing engine on Stripe and Stedi insurance APIs, splitting charges between insurer and patient with webhook-driven payment and claim updates.

Medical AI Assistant · LoRA, OCR, reranking — Freelance

- Fine-tuned an LLM with **LoRA** on neurology, orthopedics and dermatology expertise so an agent listens to consultation transcripts and checks the suggestions of junior doctors against the medical model.
- Added multimodal retrieval over X-rays, MRIs and prior reports, with OCR on previous reports and reranking of the retrieved evidence.

Rail Network GraphRAG · GraphRAG, LLM agent — Freelance

- Built a **GraphRAG** over a Polish railway management system with many cross-dependencies, so an agent answers passenger queries and books tickets.

Earlier Projects · CNN, MERN and Java — 2021 – 2023

- Medical Image Diagnosis: CNN and SVM (HOG) classifiers to detect tumors in MRI and CT scans, served through FastAPI. Also MERN e-commerce platforms (Camille Jewellery, Paradocs) and a JavaFX inventory system.

EDUCATION

BS Computer Science · COMSATS University Islamabad

- **Gold Medalist**, 1st in batch (Dean's Award, Nov 2024), CGPA 3.7/4.0. Merit Scholarship with a full four-year tuition waiver.
- **Certifications:** Generative AI with LLMs (Coursera, 2024); Supervised Machine Learning (DeepLearning.AI, 2024).